

ECE 568: Computer Security

Course Syllabus

Spring 2025
Courtney Gibson

General Information

Welcome to ECE 568!

This course covers principles of computer systems security. We will start by examining a variety of security vulnerabilities in general software applications and provide you with the knowledge to identify these weaknesses in your own work, understand how these vulnerabilities could be exploited, and develop techniques you could use to help defend against such attacks. We will then cover the basic elements of modern cryptography, and the security of operating systems, networks, embedded (“IoT”) systems and web-/cloud-based applications.

Lectures will be delivered **in-person**, and you are highly-encouraged to attend. Opportunities during the lectures will be provided to interact, ask questions and confirm your understanding of the material.

Topics Covered

We will cover the following topics in class; slides will be posted prior to the lectures to assist your note-taking:

- **Section 1: Introduction**

- Course introduction

- Quick review: Operating Systems, assembly language, software tools

- Basic principles of computer security

- Confidentiality, Integrity, Availability

- Primary Learning Goals:*

- *Ability to explain the fundamental goals of Confidentiality, Integrity and Availability, and how they each can be applied to creating secure product designs.*
 - *Ability to use basic software debugging tools (e.g., gdb) to analyze program stack behaviour.*

- **Section 2: Software Code Vulnerabilities**

Injection-type attacks: buffer overflows, format strings, double-free, Return-Oriented Programming etc.

Library attacks: encapsulation, etc.

Defenses and good programming practice

Primary Learning Goals:

- *Ability to explain and execute the most common attacks against software vulnerabilities, in order to locate design and implementation flaws.*
- *Ability to use appropriate tools and best-practice coding techniques to protect against these types of attacks in your own work.*

- **Section 3: Cryptography**

Encryption: Symmetric (block/stream ciphers) and Public Key

Secure key exchange

Hashes and Signatures: MAC, HMAC, bcrypt, Merkle trees, TOTP/HOTP

Secure Communication Protocols: SSL, VPN

Primary Learning Goals:

- *Ability to select appropriate encryption techniques to meet both the Data Confidentiality requirements and the performance requirements of your software.*
- *Ability to securely exchange cryptographic keys over insecure network channels.*
- *Ability to use hash-based tools to provide Data Integrity and Data Authentication, through the application of MACs, HMACs and digital signatures.*

- **Section 4: Web Security**

Web authentication, cookies

Web attacks: Cross-Site Scripting (XSS), CSRF, etc.

SQL Injection

Cloud security

Primary Learning Goals:

- *Ability to safely use web cookies to secure secure sessions and end-user data.*
- *Ability to identify and demonstrate the most common web-based attacks, in order to locate design and implementation flaws.*
- *Ability to use appropriate tools and best-practice coding techniques to protect against these types of attacks in your own work.*

- **Section 5: IoT / Embedded Systems**

Access control models
Hardware Security Modules (HSMs)
Side-Channels and Covert Channels
Attacks on embedded/IoT systems

Primary Learning Goals:

- *Ability to identify potential sources of side- and covert-channels that could be used to leak information out of both hardware- and software-based systems.*
- *Ability to identify and describe potential injection vulnerabilities, and how they might compromise the security of IoT/embedded systems.*

- **Section 6: Network Security**

Network-layer security risks: BGP, ARP, DNS, TCP/IP, etc.

Primary Learning Goals:

- *Ability to identify how vulnerabilities in many standard networking protocols could be exploited by a third-party to pose risks to confidentiality, integrity and availability of your software systems.*
- *Ability to use appropriate tools and best-practice techniques to protect your own systems and designs.*

- **Section 7: Malware**

Viruses and worms
Rootkits and bootkits

Primary Learning Goals:

- *Ability to identify the mechanisms through which computer viruses, worms, rootkits and bootkits function and propagate, and the risks these may pose to your systems.*

- **Section 8: Physical Security**

Physical security system design and vulnerabilities

Primary Learning Goal:

- *Ability to explain the most common vulnerabilities and defenses in physical security systems, and relate this to design challenges in digital security designs.*

Graduate Attributes:

- 4A. Design: Demonstrate ability to frame a complex, open-ended problem in engineering term. [Developed]
- 5A. Use of Engineering Tools: Demonstrate ability to use fundamental modern techniques, resources and engineering tools. [Developed]

- 5B. Use of Engineering Tools: Demonstrate ability to use discipline specific techniques, resources and engineering tools. [Developed]
- 6C. Individual and Team Work: Demonstrate success in a team based project. [Developed]
- 10C. Ethics and Equity: Demonstrate the ability to act ethically and demonstrate individual accountability. [Developed]

Instructor

Prof. Courtney Gibson, P.Eng.
 Department of Electrical and Computer Engineering
 courtney.gibson@utoronto.ca
Office hours by appointment

Timetable

The timetable for the course is shown below. You are highly encouraged to attend the weekly lecture (three hours) each week; recordings will be posted after the lecture for those unable to attend. The Teaching Assistants will be on-line and answering questions live during the designated lab times, and will be answering questions asynchronously through the course forums outside of those hours.

Lecture	Monday (L0101)	6:10pm – 9:00pm MY 150
Labs	Tuesday (P0101)	5pm – 6pm (ATTENDANCE IS OPTIONAL) Online (or SF1012)
	Friday (P0102)	12pm – 3pm (ATTENDANCE IS OPTIONAL) Online (or SF1012)
	Friday (P0103)	9am – 12pm (ATTENDANCE IS OPTIONAL) Online (or SF1012)
	Tuesday (P0104)	9am – 12pm (ATTENDANCE IS OPTIONAL) Online (or SF1012)
	Monday (P0105)	9am – 12pm (ATTENDANCE IS OPTIONAL) Online (or SF1012)

Course Website

Information on ECE568, including important announcements, lecture slides, recordings, assignments and course marks can be found on Quercus (<https://q.utoronto.ca/>). Please visit the website on a

regular basis for up-to-date information.

Please note that any grades posted are for your information only, so you can view and track your progress through the course. No grades are considered official until they have been formally approved and posted on ACORN at the end of the course. Please contact me as soon as possible if you think there is an error in any grade posted on Quercus.

Textbook

There is no required textbook for the course. However, the instructor will provide a list of reference books in class and lecture slides on the course web site. Also, various other resources will be available on the web site.

Labs

The labs consist of a number of programming exercises that will take a substantial amount of your time – but students who have taken this course in past years consistently report that these labs have taught them a tremendous amount of useful material. Please take them seriously, and ask for help early if you need it.

>>> IMPORTANT: Some of the lab assignments will have us doing things that we are not normally “supposed” to be able to do (e.g., attacking applications with invalid input, exploiting a vulnerability, and changing how they run). The nature of what we are doing means that different operating systems, different processors and different system security policies may impact how our “attacks” run. The TAs will mark your lab on the ECF lab workstations (p____.ecf.utoronto.ca), **not** any other of the ECF machines). Please make sure you are doing your work on the correct machines.

For those labs that require a compiler, you should be using the C compiler installed in `/usr/bin/gcc`. You may do the labs on your own machines – but, if you do, it is your responsibility to make sure that your solutions work on the ECF computers. My suggestion is to SSH into the ECF machines from home, if you prefer to work remotely... rather than discovering at the last minute that your personal operating system or computer configuration is different than the lab machines. If you do not have an ECF account, please email me and I will sort things out for you.

The TAs will be using some automated scripts to aid them in grading the labs: as a result, it is important that you follow the submission instructions for each lab carefully. You are encouraged to include documentation for your labs (not exceeding 1 page, please no essays!). If your labs do not work completely, the TAs may use this documentation to assign part marks. Lab attendance is not required: your lab sections are simply assigned to give you time in your schedule, and a place to work, if you require it. **Labs may be done individually or in groups of two students.**

Tutorials

There are no tutorials in this course. Please make use of the TAs and office hours with the instructor.

Notice of content sharing

Course videos, lab handouts, slides, any provided code, and other course materials belong to the instructor, and are protected by copyright. **You are permitted to download session videos and materials for your own academic use, but you should not copy, share, or use them for any other purpose without the explicit permission of the instructor. Posting of provided code onto public file- or code-sharing sites is specifically prohibited.**

Course Policies

All students, faculty and staff are expected to follow the University's guidelines and policies on academic integrity. For students, this means following the standards of academic honesty when writing assignments, collaborating with fellow students, and writing tests and exams. Ensure that the work you submit for grading represents your own honest efforts. Plagiarism – representing someone else's work as your own or submitting work that you have previously submitted for marks in another class or program – is a serious offence that can result in sanctions. You and your lab partner are jointly responsible for ensuring that your submitted lab work is original work. Speak to me or your TA for advice on anything that you find unclear. Consult the [Code of Behaviour on Academic Matters](#) for a complete outline of the University's policy and expectations. For more information, please see the [U of T Academic Integrity website](#).

If for some valid reason you are unable to submit the lab on time, please provide an explanation and appropriate documentation.

Re-grading

Everybody makes mistakes, including TAs and myself! I will ensure you receive fair evaluation for your work in this course. If you feel that there has been a grading mistake, you can request a reevaluation of your work within three weeks of the test or lab results being returned. You should submit a short note explaining which questions are in error and why you think you deserve a regrade. A TA or the instructor will regrade the entire assignment, and fix any marking mistakes. (It is not guaranteed that you will receive a higher mark: it's unlikely, but you may end up with a lower grade on your assignment if it's determined that the marking key was not properly applied.)

Important Dates

The release date and due dates for the labs and assignments are shown below. A handout for each lab will be available from the course web site, and no hard copies will be provided in class. **Labs are due by no later than 11:59pm EST/EDT on Friday of the week they are due.**

Week 1 - Jan 6-10	Course orientation: no labs.
Week 2 - Jan 13-17	Lab #1 released.
Week 3 - Jan 20-24	Lab hours begin.
Week 4 - Jan 27-31	Lab #1 due. Lab #2 released.
Week 5 - Feb 3-7	

Week 6 - Feb 10-14	Lab #2 due.
Feb 17-21	Reading Week: No lectures or labs.
Week 7 - Feb 24-28	Midterm. No lecture. Lab #3 released.
Week 8 - Mar 3-7	
Week 9 - Mar 10-14	Lab #3 due.
Week 10 - Mar 17-21	Lab #4 released.
Week 11 - Mar 24-28	
Week 12 - Mar 31- Apr 4	Lab #4 due.
Week 13 - Apr 7-11	Last week: final exam review.

Marking and Evaluation

There will be one mid-term exam during the course, and a final exam during the final exam period.

For both exams, you are permitted to bring a single **aid-sheet**; it can be double-sided, no larger than 8.5x11", and may be computer-printed or hand-written. The contents of the aid sheet must be your own work.

The **midterm exam** will be 90 minutes in length, and will cover the material from the lectures and labs from the first half of the course. It includes a number of short-answers and short-essay questions. The details for the exact timing of the mid-term will be provided in class and on the web site; it will *likely* take place during our regularly-scheduled lecture time on Monday, February 24th.

The **final exam** will be 2.5 hours in length, and will cover the material from all lectures and labs. It includes a number of short-answer and short-essay questions. The final exam is a required deliverable for this course. The details of final exam timing and delivery is controlled by the Department (I have no control over when/where the exam will be scheduled); details will be provided closer to the exam date.

Copies of previous midterms and final exams will be provided on Quercus, as a study aid.

The composition of the final mark is as follows:

Labs:	25%
Mid-Term Exam:	35% (90 minute exam)
Final Exam:	40% (2.5 hour exam)
Calculator Type:	4 (none)
Exam Type:	C (single reference sheet, both sides)
Credit Value:	0.5

University Land Acknowledgement

I wish to acknowledge this land on which the University of Toronto operates. For thousands of years, it has been the traditional land of the Huron-Wendat, the Seneca, and the Mississaugas of the Credit. Today, this meeting place is still the home to many Indigenous people from across Turtle Island and we are grateful to have the opportunity to work on this land. Learn more about Canada's relationship with Indigenous Peoples [here](#).

Indigenous Students' Supports

If you are an Indigenous engineering student, you are invited to join a private Discord channel to meet other Indigenous students, professors, and staff, chat about scholarships, awards, work opportunities, Indigenous-related events, and receive mentorship. Email [Professor Bazylak](#) or [Darlee Gerrard](#) if you are interested.

Indigenous students at U of T are also invited to visit First Nations House's (FNH) Indigenous Student Services for culturally relevant programs and services. If you want more information on how to apply for Indigenous specific funding opportunities, cultural programs, traditional medicines, academic support, monthly social events or receive the weekly newsletter, go to the FNH [website](#), [email](#) or follow FNH on social media: [Facebook](#), [Instagram](#), or [TikTok](#). A full event calendar is on the CLNX platform. Check CLNX often to see what new events are added!

Wellness and Mental Health Support

As a university student, you may experience a range of health and/or mental health challenges that could result in significant barriers to achieving your personal and academic goals. The University of Toronto and the Faculty of Applied Science & Engineering offer a wide range of free and confidential services that could assist you during these times.

As a U of T Engineering student, you have a Departmental [Undergraduate Advisor](#) or a Departmental [Graduate Administrator](#) who can support you by advising on personal matters that impact your academics. Other resources that you may find helpful are listed on the [U of T Engineering Mental Health & Wellness webpage](#), and a small selection are also included here:

-  [U of T Engineering's Mental Health Programs Officer](#)
-  [Accessibility Services](#) & the [On-Location Advisor](#)
-  [Health & Wellness](#) and the [On-Location Health & Wellness Engineering Counsellor](#)
-  [Graduate Engineering Council of Students' Mental Wellness Commission](#)
-  [SKULE Mental Wellness](#)
-  [U of T Engineering's Learning Strategist](#) and [Academic Success](#)
-  [Registrar's Office](#) and [Scholarships & Financial Aid Office & Advisor](#)

We encourage you to access these resources as soon as you feel you need support; no issue is too small. If you find yourself feeling distressed and in need of more immediate support, consider reaching out to the counsellors at [U of T Telus Health Student Support](#) or visiting U of T Engineering's [Urgent Support – Talk to Someone Right Now](#).

Accommodations

The University of Toronto supports accommodations for students with diverse learning needs, which may be associated with mental health conditions, learning disabilities, autism spectrum, ADHD, mobility impairments, functional/fine motor impairments, concussion or head injury, visual impairments, chronic health conditions, addictions, D/deaf, deafened or hard of hearing, communication disorders and/or temporary disabilities, such as fractures and severe sprains, or recovery from an operation.

If you have a learning need requiring an accommodation the University of Toronto recommends that students [register with Accessibility Services](#) as soon as possible.

We know that many students may be hesitant to reach out to Accessibility Services for accommodations. The purpose of academic accommodations is to support students in accessing their academics by helping to remove unfair disadvantages. We can assess your situation, develop an accommodation plan with you, and support you in requesting accommodation for your course work. The process of accommodation is private; we will not share details of your needs or condition with any instructor.

If you feel hesitant to register with us, we encourage you to reach out for further information and resources on how we can support. It may feel difficult to ask for help, but it can make all the difference during your time here.

Phone: 416-978-8060

mail: accessibility.services@utoronto.ca

Equity, Diversity and Inclusion

Looking for community? Feeling isolated? Not being understood or heard?

You are not alone. You can talk to anyone in the Faculty that you feel comfortable approaching, anytime – professors, instructors, teaching assistants, [first-year](#) or [upper years](#) academic advisors, student leaders or the [Assistant Dean of Diversity, Inclusion and Professionalism](#).

You belong here. In this class, the participation and perspectives of everyone is invited and encouraged. The broad range of identities and the intersections of those identities are valued and create an inclusive team environment that will help you achieve academic success. You can read the evidence for this approach [here](#).

You have rights. The [University Code of Student Conduct](#) and the [Ontario Human Rights Code](#) protect you against all forms of harassment or discrimination, including but not limited to acts of racism, sexism, Islamophobia, antisemitism, homophobia, transphobia, ableism, classism and ageism. Engineering denounces unprofessionalism or intolerance in language, actions or interactions, in person or online, on- or off-campus. Engineering takes these concerns extremely seriously and you can confidentially disclose directly to the Assistant Dean for help [here](#).

Resource List:

-  [Engineering Equity, Diversity & Inclusion Groups, Initiatives & Student Resources](#)
-  [Engineering Positive Space Resources](#)

- Request a religious-based accommodation [here](#)
- Email Marisa Sterling, P.Eng, the Assistant Dean, Diversity, Inclusion & Professionalism [here](#)
- Make a confidential disclosure of harassment, discrimination or unprofessionalism [here](#) or email engineering@utoronto.ca or call 416.946.3986
- Email the Engineering Society Equity & Inclusivity Director [here](#)
- [U of T Equity Offices & First Nations House Resources](#)